

Q1. DIRECTIONS for questions 1 to 3: Select the correct alternative from the given choices.

If α_1 and α_2 are the real roots of $x^2 - px + 12 = 0$, then which of the following statements is definitely true?

☐ a) $|\alpha_1 + \alpha_2| \leq 2\sqrt{3}$

☐ b) $|\alpha_1 - \alpha_2| \leq 2\sqrt{3}$

☐ c) $|\alpha_1 + \alpha_2| \geq 4\sqrt{3}$

☐ d) $|\alpha_1 - \alpha_2| \geq 4\sqrt{3}$

Q2. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

Let Q be a recurring decimal of the form $Q = 0.P_1P_2P_3P_1P_2P_3 \dots$, where P_1, P_2, P_3 are single digits, out of which at most one is zero. Which of the following numbers when multiplied by Q , necessarily gives an integer?

☐ a) 1998

☐ b) 198

☐ c) 1728

☐ d) 1988

Q3. DIRECTIONS *for questions 1 to 3:* Select the correct alternative from the given choices.

A rectangular floor of dimensions 170 cm by 140 cm is to be covered (not necessarily completely) by rectangular tiles of size 90 cm by 40 cm. While the tiles may be placed in any direction, their edges have to be parallel to the edges of the floor. No tile should either overlap another tile or overshoot any edge of the floor. What is the maximum number of tiles that can be accommodated on the floor?

☒ a) 5 **✗ Your answer is incorrect**

☐ b) 7

☐ c) 4

☐ d) 6

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

If $25^{\{\log_5 12 + 3 \log_x 16\}} = \frac{9}{256}$, find the value of x .

Q5. DIRECTIONS for questions 5 and 6: Select the correct alternative from the given choices.

A piece of work can be done by 11 men and 16 boys in 2 days. If the same work can be done by 5 men and 11 boys in 4 days, in how many days can 1 man and 4 boys complete the same work?

☐ a) $16\frac{2}{5}$

☐ b) $20\frac{1}{2}$

☐ c) $41\frac{1}{4}$

☐ d) $14\frac{3}{5}$

Q6. DIRECTIONS *for questions 5 and 6:* Select the correct alternative from the given choices.

If $\theta \in [0, 2\pi]$, for which of the following ranges of θ is $|\sin\theta + \cos\theta| < \frac{1}{\sqrt{2}}$?

☐ a) $\left(\frac{\pi}{4}, \frac{5\pi}{12}\right)$

☐ b) $\left(\frac{2\pi}{3}, \frac{11\pi}{12}\right)$

☐ c) $\left(0, \frac{\pi}{6}\right)$

☐ d) $\left(\frac{7\pi}{6}, \frac{3\pi}{2}\right)$

Q7. DIRECTIONS *for question 7:* Type in your answer in the input box provided below the question.

A milkman purchased milk at a certain price, diluted it by adding water and then sold the mixture at $33\frac{1}{3}\%$ above his cost price and thereby made a profit of 40%. Find the quantity (in ml) of water he added for every litre of milk purchased.

Q8. DIRECTIONS *for question 8:* Select the correct alternative from the given choices.

Two trains started simultaneously, at 11:00 am, from Delhi and Jaipur, moving towards Jaipur and Delhi respectively, and crossed each other at 3:00 pm. If the train starting from Jaipur reached its destination 88 minutes before the other train, when did the train starting from Delhi reach Jaipur?

☐ a) 7:48 pm

☐ b) 7:30 pm

☐ c) 7:20 pm

☐ d) 8:00 pm

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

Akhil has a board, similar to a chessboard, except that it has only 6×6 squares. He calculated the number of rectangles (including squares) on the board. How many distinct values can the area of these rectangles assume?

Q10. DIRECTIONS *for questions 10 and 11:* Select the correct alternative from the given choices.

Twenty identical billiard balls (spherical) are perfectly arranged on a table in four layers, with the bottom most through the topmost layers comprising 10, 6, 3 and 1 balls respectively, such that the resulting shape resembles a regular tetrahedron. If the radius of each ball is R , what is the height of the topmost point of the formation from the surface of the table?

- ☐ a) $2(2 + \sqrt{3})R$
- ☐ b) $2(1 + \sqrt{3})R$
- ☐ c) $2(1 + \sqrt{6})R$
- ☐ d) $2(1 + 2\sqrt{3})R$

Q11. DIRECTIONS *for questions 10 and 11: Select the correct alternative from the given choices.*

Mohan subtracted a certain quantity x from each of 50, 80 and 130. If he found that the resulting numbers formed a geometric progression, find the common ratio of the geometric progression.

- ☐ a) $\frac{3}{2}$
- ☒ b) $\frac{5}{3}$ ✓ **Your answer is correct**
- ☐ c) **2**
- ☐ d) $\frac{8}{5}$

Q12. DIRECTIONS *for question 12:* Type in your answer in the input box provided below the question.

Two non-zero real numbers, x and y , satisfy the equations given below:

$$x + y + x^2 + 10y^2 = 0$$

$$\frac{x}{y} + \frac{y}{x} = 9$$

If $x + y \neq 0$, find the value of $27y$.

Q13. DIRECTIONS for questions 13 to 15: Select the correct alternative from the given choices.

$$(22)_3 + (22)_4 + (22)_5 + \dots + (22)_n =$$

- ☐ a) $3n^2 - 6n - 1$
- ☐ b) $2n^2 - 3n - 1$
- ☐ c) $n^2 + 3n - 10$
- ☐ d) $n^3 - 3n^2 + 3n + 2$

Q14. DIRECTIONS *for questions 13 to 15:* Select the correct alternative from the given choices.

If $\left(\sqrt[3]{\frac{16}{81}}\right)^{2x-5} = \left(\frac{27}{8}\right)^{x-11}$, then the value of x is equal to

- ☐ a) 3.
- ☒ b) 7. ✓ Your answer is correct
- ☐ c) 5.
- ☐ d) 6.

Q15. DIRECTIONS *for questions 13 to 15:* Select the correct alternative from the given choices.

A and B are playing a game, where A throws a die and B pays him the amount (in rupees) equal to the number (from 1 to 6) that turns up on the die. If for every throw A has to pay ₹3 to B, what is the expected net amount earned by A in 300 throws?

- ☐ a) ₹150
- ☐ b) ₹100
- ☐ c) ₹50
- ☐ d) ₹300

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

If $f(x.y) = f(x).f(y)$, where $x > 0$, $y > 0$, and $f(128) = 8$, find the value of $f(1).f(2).f(4).f(8).f(16).f(32).f(64)$.

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.

A number when divided by 31 leaves a remainder of 3. If the same number when divided by 93 leaves a remainder of r , how many possible values can r assume?

☐ a) 1

☐ b) 2

☐ c) 5

☐ d) 3

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

How many eight-digit numbers are there for which the sum of the digits is 4?

Q19. DIRECTIONS *for question 19:* Select the correct alternative from the given choices.

If the simple interest and the compound interest on a certain sum of money, for two years, at the same rate of interest, are ₹720 and ₹792 respectively, find the rate of interest.

☐ a) 5%

☐ b) 10%

☐ c) 20%

☐ d) 36%

Q20. DIRECTIONS *for questions 20 and 21:* Type in your answer in the input box provided below the question.

Find the number of trailing zeroes in $^{100}C_{50}$.

Q21. DIRECTIONS *for questions 20 and 21:* Type in your answer in the input box provided below the question.

Let S_m denote the sum of the first m natural numbers. For how many values of m less than or equal to 100, is S_m divisible by 15?

Q22. DIRECTIONS *for question 22: Select the correct alternative from the given choices.*

The average weight of the students in a class is 45 kg. If a group of 15 students, with an average weight of 60 kg, joins the class, the average weight of the entire class increases to 50 kg. What is the initial number of students in the class?

- ☐ a) 24
- ☐ b) 20
- ☐ c) 25
- ☒ d) 30 ✓ **Your answer is correct**